

Computing power needed for running ChatGPT?

Posted by u/stimmen • 0 points • 15 comments

Hey folks, I didn't manage to find information how much computing power OpenAI needs for running ChatGPT. Does anyone have a clue or can point me to relevant sources?

After all I'm interested in this question: Is the resource demand for running this system extraordinary huge - or not so much? I'd like to get an idea. Something like "an average question to ChatGPT needs x kWh" or the like.

Comments

u/Prize_Bus_132 • 1 points • 2024-04-27 20:08 UTC

The question is about COMPUTER power, i.e., the processing power required. Some folks mistaking it for electricity power. Unfortunately I don't know the answer to question

u/r2bl3nd • 0 points • 2023-02-28 18:54 UTC

GPT-2 used 1.5 billion parameters and was publicly released, so if you research, I'm sure you can find hard data for the amount of computing power. You could compare it to the power required by the original GPT model (117 million parameters) and see if it's a linear scaling amount or not.

GPT-3 uses 175 billion parameters, so given what one can find about GPT and GPT-2, surely we can extrapolate and figure out what GPT-3 needs.

u/OppressorOppressed • 1 points • 2023-05-06 19:40 UTC

I can run GPT-2 on my machine, and its an old desktop

u/[deleted] • 1 points • 2023-02-28 17:39 UTC

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u/Virtualcosmos • 2 points • 2023-02-28 14:45 UTC

It told me that its normal version (not learning of course) only need a normal high end pc from today, aka an i7 from the 10 series, 32 gb of ram and a RTX 3080 graphic card. It was mindblowing for me to think I could have all the knowledge this AI has in my pc without even internet connection. I wish I could have it

u/NickyYeet • 1 points • 2023-03-11 18:36 UTC

Not sure where you got this from, chatgpt requires a lot more processing power than just a 3080, people say that it requires 400gb of vram (pre-trained model) while others say for it to be trained you'll need about 800gb of vram. All of this needs to be loaded into gpu memory otherwise it would probably take days to generate some usable text.

u/CaptainCremin • 1 points • 2023-08-22 00:27 UTC

Not to drag out an old thread, but I don't think this can be right. If that was true how could so many instances of ChatGPT run at once?

u/NickyYeet • 1 points • 2023-08-22 14:49 UTC

This is pretty old but the answer is money. Lots and lots of money. ChatGPT has support from billionaires, and as they've been gaining popularity, Microsoft partnered with them to use Microsoft azure servers to run their ai

u/CaptainCremin • 1 points • 2023-08-23 03:51 UTC

I would've thought it was cost prohibitive, but I guess people just have stupid amounts of money.

Do you know how much it costs to run an AI in it's learning mode?

u/NickyYeet • 1 points • 2023-08-23 23:41 UTC

A quick lookup shows that GPT-3 took about 1.4 million dollars to train and GPT-4 took about 63 million to train. And according to a company analysis by Dylan Patel, ChatGPT can take up to 700,000 dollars a day to run (take this with a grain of salt - values can always change).

u/Virtualcosmos • 1 points • 2023-03-12 06:51 UTC

hmm I think you are thinking in the learning part of the model, where it needs a cluster computer to learn from data. What I asked chatGPT is what is required to run its model already trained, as it's right now

u/NickyYeet • 1 points • 2023-03-12 18:12 UTC

As it is right now even an already trained model still would require much more than a consumer/enthusiast gpu. If you look at BLOOM the biggest competitor(still not as good as chatGPT) still requires about 4-8 nvidia a100 gpus to run a copy. There's a common misconception that prelearned instructions are easier to deal with. They are, but not by a lot.

u/Virtualcosmos • 1 points • 2023-03-12 19:51 UTC

if it as you say, how damn many parameters has that neural network?

u/NickyYeet • 1 points • 2023-03-13 00:59 UTC

176 billion parameters, 1 billion more than chatgpt according to google, but chatgpt is constantly growing.

u/stimmen • 5 points • 2023-02-28 16:10 UTC

Probably in the not so distant future.

u/eras • 1 points • 2023-02-28 10:16 UTC

I read from some article(TM) that the nodes they have at Azure take around 3 kilowatts. So based on that and the duration it takes to respond you could come up with some number.

Those devices had like 4 (or 6?) GPUs though and I'm not sure if all of them are used for constructing a single response (and I don't know if it might be slower due to other requests on the same GPU even at quiet times, or if the performance of free tier is throttled instead of just de-prioritized), though, so it might be trickier to come up with an exact number. But maybe ballpark?

u/[deleted] • -1 points • 2023-02-28 10:14 UTC

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u/Extreme-Snow-5888 • 1 points • 2023-02-28 09:57 UTC

I asked Bing AI your question. It was very unhelpful. Bing clearly sucks.

> ChatGPT is a chatbot that is fine-tuned from GPT-3.5, a language model trained by OpenAI¹. ChatGPT was optimized for dialogue by using reinforcement learning with human feedback². It can answer questions on various topics and domains, as well as generate natural and lifelike responses¹.

> However, ChatGPT also requires a lot of computing power and energy for its training and operation. According to one report³, just to develop training models and inferencing alone for ChatGPT can require 10,000 Nvidia GPUs and probably more. This would be a steep investment for cloud providers and organizations alike.

> According to one source¹, a 3-billion parameter model can generate a token in about 6ms on an A100 GPU. If we scale that up to the size of ChatGPT, which has 175 billion parameters², it should take 350ms for an A100 GPU to print out a single word. The cost of running ChatGPT is estimated at \$100K per day, or \$3M per month.

> To run ChatGPT, you need access to OpenAI's API³, which provides a cloud-based platform for interacting with various models, including ChatGPT. You can sign up for an account and request access to the API here: <https://openai.com/api/>

> According to ChatGPT itself¹, users need a server or virtual machine with at least 16 GB of RAM, a CPU with at least four cores, and a modern GPU with at least 8 GB memory to host ChatGPT in a data center and develop AI-based applications.

> However, another source² suggests that users need a high-end CPU with at least 16 cores, at least 64 GB of RAM, a high-end GPU with at least 16 GB of VRAM, and a large amount of storage space (at least several hundred GB) to use ChatGPT effectively.

> The system requirements may vary depending on the specific use case and model configuration of ChatGPT. I hope this helps you understand more about ChatGPT. Do you have any other questions?

I doubt that this is accurate though. I read somewhere that Chat GPT requires 800GB of RAM to run.